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22PH102 – ELECTROMAGNETISM AND MODERN PHYSICS

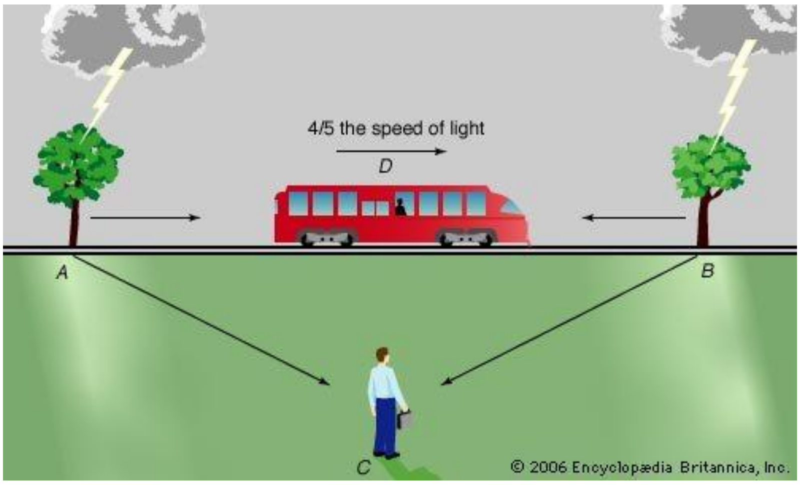
Unit-IV: MODERN PHYSICS_LM2

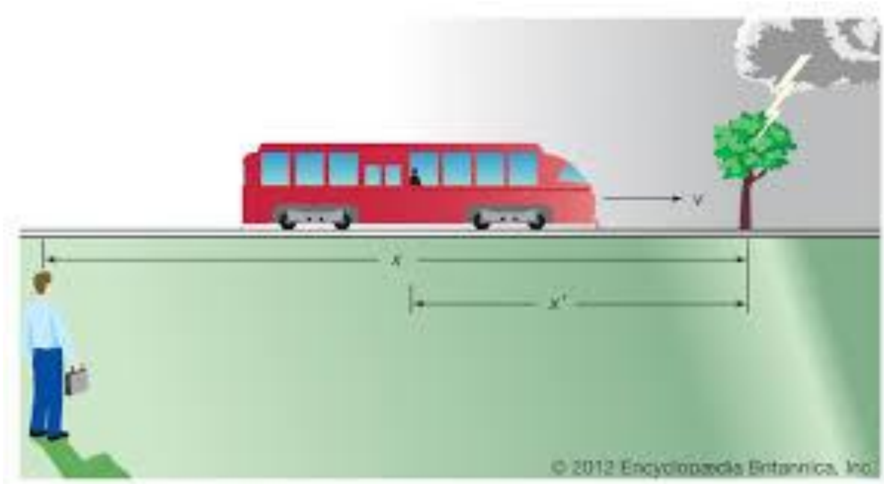
Consequences of the Special Theory ofRelativity

1. Simultaneity the Relativity ofTime
2. TimeDilation
3. LengthContraction
4. Relativistic Energy

Relativity of simultaneity

It is the concept that distant simultaneity whether two spatially separated events occur at the same time is not absolute, but depends on the observer's reference frame.





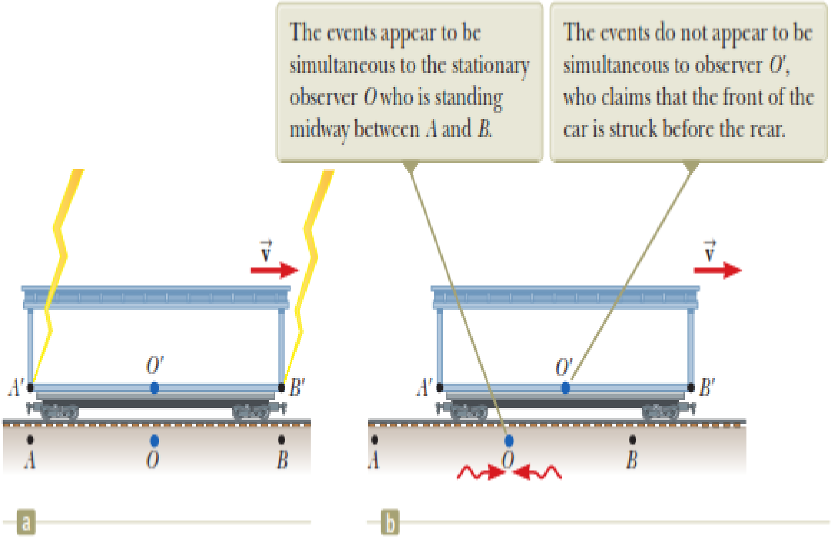
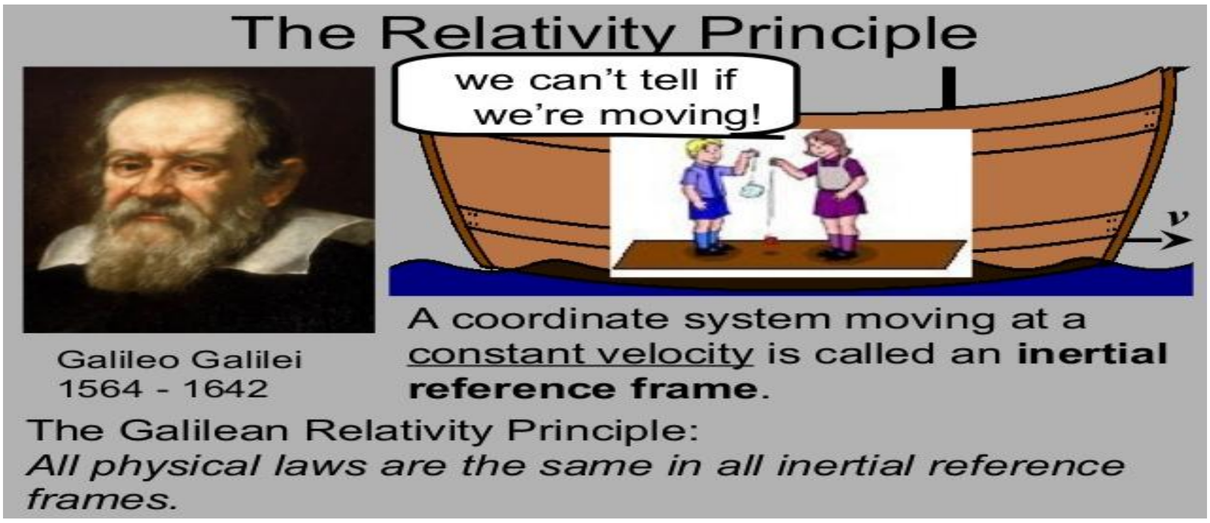


Figure 39.5 (a) Two lightning bolts strike the ends of a moving boxcar. (b) The leftward-traveling light signal has already passed O', but the rightward-traveling signal has not yet reached O'.

Two events are defined to be simultaneous if an observer measures them as occurring at the same time (such as by receiving light from the events). Two events are not necessarily simultaneous to all observers. In short, Einstein realized, simultaneity is what's relative.



Simultaneity and Time Dilation

Introduction:

Simultaneity and time dilation are two fundamental concepts in Einstein's Special Theory of Relativity. They challenge our common-sense notions of time and how events occur simultaneously.

How can we say if two events are simultaneous?

–They occur at the same time!

If two events are simultaneous in one frame, are they simultaneous in others?

–Newtonian: YES! (absolute time)

–Relativistic: NO! (no absolute time)

1. Simultaneity:

Simultaneity refers to the concept of events occurring at the same time from the perspective of an observer. However, in the context of Special Relativity, simultaneity is relative and depends on the observer's frame of reference.

Galilean Relativity vs. Einsteinian Relativity: In Galilean relativity, events that are simultaneous in one frame of reference are simultaneous in all frames. However, Einstein proposed a different understanding in Special Relativity.

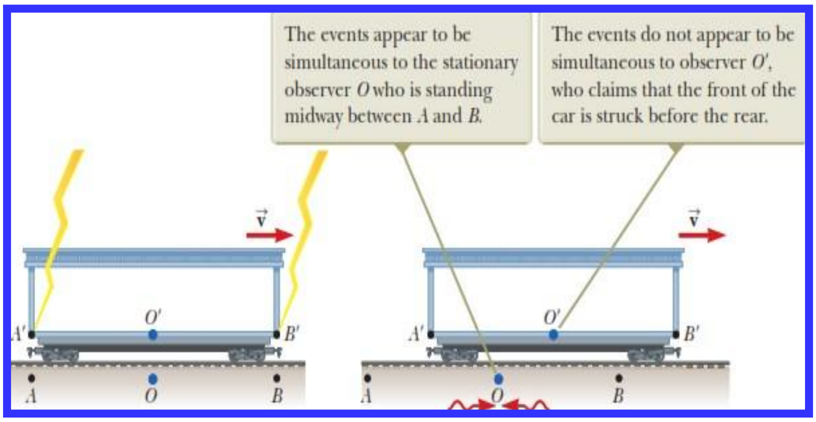
Relativity of Simultaneity: According to Special Relativity, two events that appear simultaneous to one observer may not appear simultaneous to another observer in relative motion. This is due to the finite speed of light and the relativity of time.

Simultaneity and the Relativity of Time

In NewtonianMechanics

If two events are simultaneous for an observer, then they are simultaneous for all observers.

Two llightning bolts striketends of a moving boxcar



For ObserverO

Two events appeartobesimultaneous

For Observer O'

Two events are not simultaneous SignalfromB' will reach first and the signal from A' reacheslater

Buttwoobservers must find thatlight travels at the samespeed

2. Time Dilation:

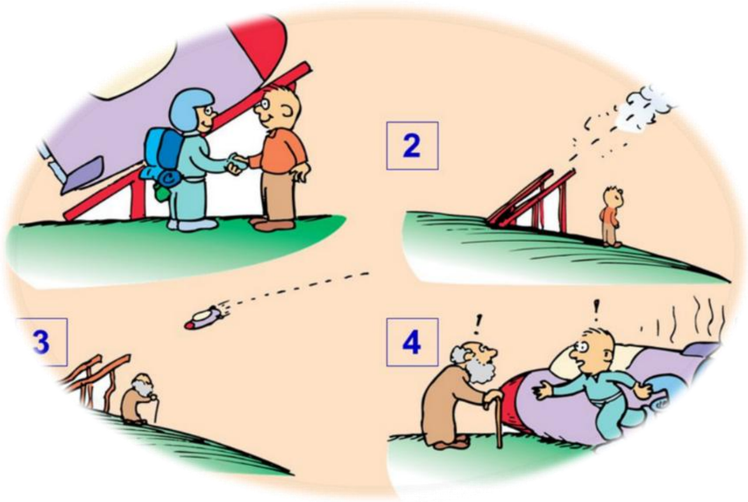
Time dilation is a phenomenon predicted by Special Relativity, stating that time appears to pass slower for an observer in motion relative to another observer at rest. It arises from the constancy of the speed of light and the relativity of simultaneity.

Implications: Time dilation leads to several counterintuitive consequences, such as the famous Twin Paradox and the slowing down of clocks in high-speed particle accelerators.

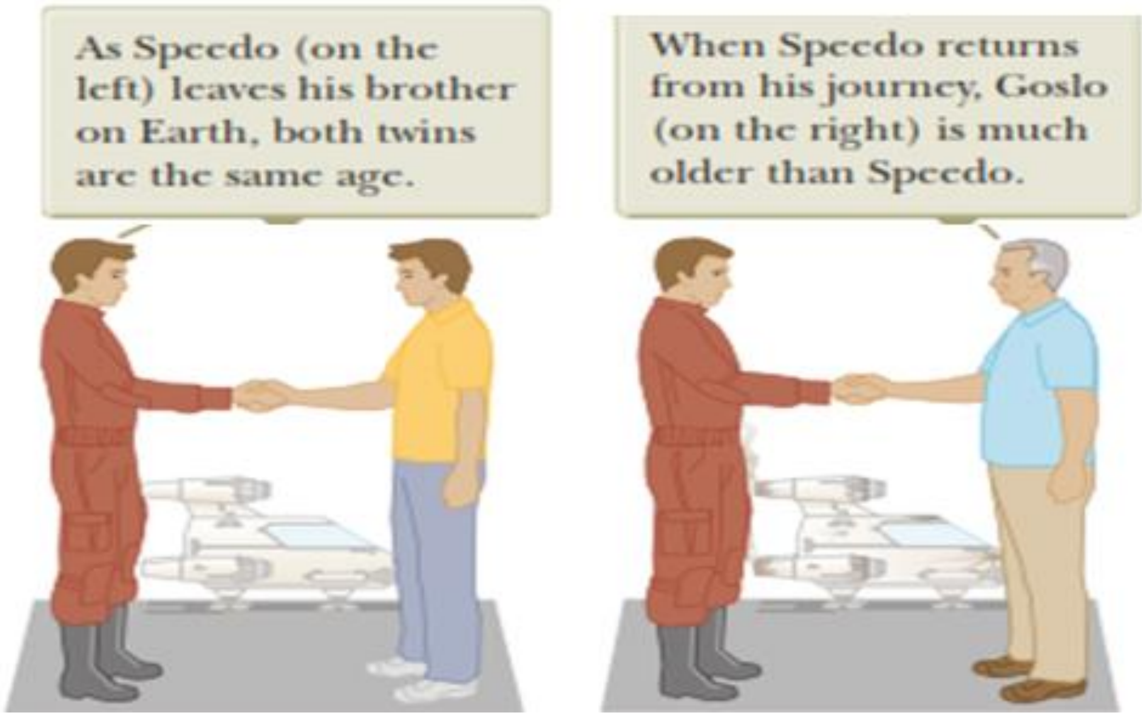
Example: Twin Paradox

Consider two twins, Alice and Bob. Alice stays on Earth while Bob travels into space at relativistic speeds and then returns. From Alice's perspective, Bob's clock appears to run slower due to time dilation. When Bob returns, he will be younger than Alice.

CONCEPT OF TWIN PARADOX



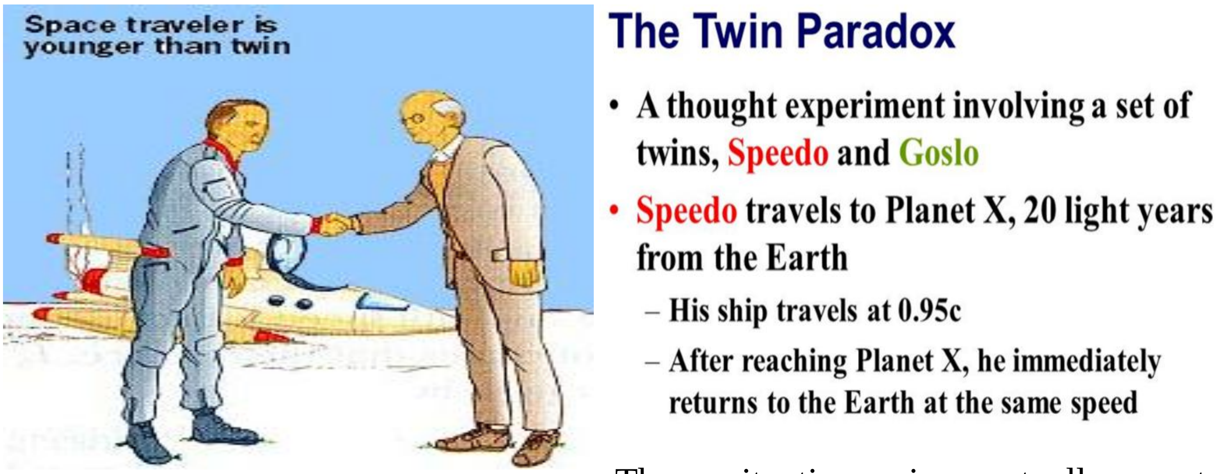
This leads to one of the strangest results of relativity – the twin paradox, which says that if one of a pair of twins makes a trip into space on a high-speed rocket, they will return to Earth to find their twin has aged faster than they have.



Speedo takes a journey to a star 20 light-years away and returns to the Earth.

Is this real? Would one twin really be younger?

- **When Speedo returns, he has aged ? years, but Goslo has aged ?? years**



The situation is actually not symmetrical.

Consider a third observer moving at a constant speed relative to Goslo.

According to the third observer, Goslo never changes inertial frames. Goslo's speed relative to the third observer is always the same.

The third observer notes, however, that Speedo accelerates during his journey When he slows down and starts moving back toward the Earth, *changing reference frames in the process.*

From The third observer's perspective, there is something very different about the motion of Goslo when compared to Speedo.

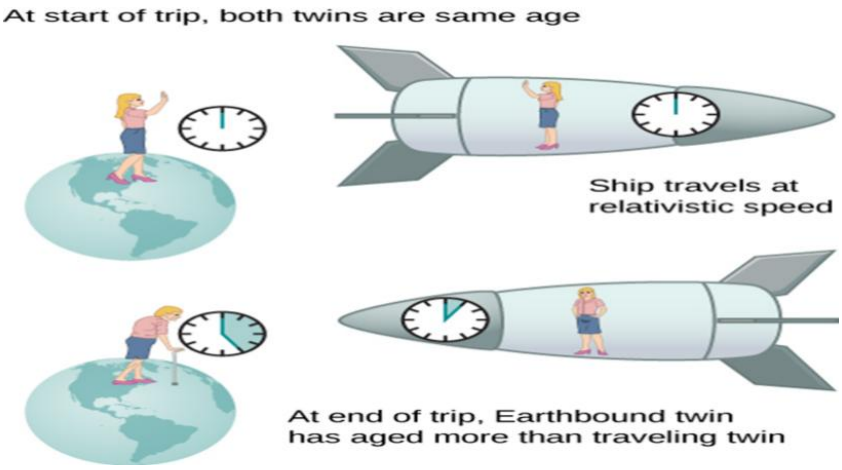
$$\Delta t = \gamma \Delta t_0$$

$$\text{where } \gamma \equiv \frac{1}{\sqrt{1-\beta^2}}$$

Δt : Time on Earth

Δt_0 : Time on spaceship

Therefore, there is no paradox: only Goslo, who is always in a single inertial frame, can make correct predictions based on special relativity.



Example: Light Clocks

Consider a light clock, consisting of two parallel mirrors with a beam of light bouncing between them. According to classical physics, the time for light to travel up and down should be the same in all reference frames. However, in Special Relativity, due to the relativity of simultaneity, observers in relative motion will disagree on whether the light reaches the top and bottom mirrors simultaneously. This disagreement leads to time dilation effects.

Example: GPS Satellites

Global Positioning System (GPS) satellites orbit Earth at high speeds. As discussed earlier, due to their motion relative to observers on Earth's surface, the clocks on these satellites experience time dilation. Without accounting for this effect, the GPS system would accumulate significant errors, rendering it inaccurate for navigation.